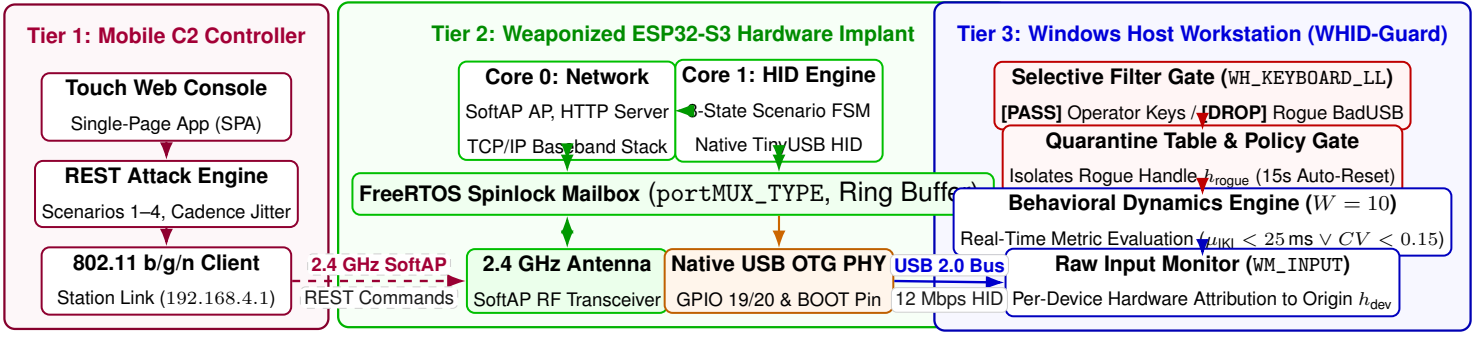

Tier 1: Mobile C2 Controller
Touch Web Console
Single-Page App (SPA)
REST Attack Engine
Scenarios 1–4, Cadence Jitter
802.11 b/g/n Client
Station Link (192.168.4.1)
2.4 GHz SoftAP
REST Commands
Tier 2: Weaponized ESP32-S3 Hardware Implant
Core 0: Network
SoftAP AP, HTTP Server
TCP/IP Baseband Stack
Core 1: HID Engine
8-State Scenario FSM
Native TinyUSB HID
FreeRTOS Spinlock Mailbox (portMUX_TYPE, Ring Buffe
2.4 GHz Antenna
SoftAP RF Transceiver
Native USB OTG PHY
GPIO 19/20 & BOOT Pin
USB 2.0 Bus
12 Mbps HID
Tier 3: Windows Host Workstation (WHID-Guard)
Selective Filter Gate (WH_KEYBOARD_LL)
[PASS] Operator Keys / [DROP] Rogue BadUSB
Quarantine Table & Policy Gate
Isolates Rogue Handle h_{rogue} (15s Auto-Reset)
Behavioral Dynamics Engine ($W = 10$)
Real-Time Metric Evaluation ($\mu_{|K|} < 25\,\text{ms} \vee CV < 0.15$)
Raw Input Monitor (WM_INPUT)
Per-Device Hardware Attribution to Origin h_{dev}